

REEM V2

Inventory Intelligence

Pharmacy Supply Chain Intelligence

Research and Development Framework

From Inventory State to Management Intelligence

September 2026

Synthetic Development Environment
Operational Use Prohibited

1 Executive Summary

REEM V2 is designed as an intelligence layer above pharmacy operational systems. Its purpose is not simply to report stock balances, but to transform operational evidence into analytical information that can support inventory decisions.

The inventory intelligence layer integrates four major evidence streams:

1. sales and demand history,
2. procurement history,
3. delivery and lead-time history,
4. current inventory state.

The current development implementation successfully integrates forecast demand with observed delivery lead times and current inventory-batch quantities.

Fifty medicines were analysed.

The current synthetic environment produced:

- 50 medicines analysed,
- 46 medicines with observed delivery lead-time evidence,
- 4 medicines using an explicit seven-day development assumption,
- no negative current-stock values,
- positive reorder points for all medicines,
- 50 medicines classified as ADEQUATE under the current development decision rule.

The result should not be interpreted as evidence that a real pharmacy has adequate stock. The dataset is explicitly synthetic and exists only for research and development.

2 Purpose

The purpose of the inventory intelligence model is to answer a management question:

How much inventory is available relative to the amount required to cover expected demand and replenishment uncertainty?

This requires the system to connect:

Demand → Lead Time → Replenishment Requirement → Current Stock → Decision.

The model therefore represents a transition from operational record-keeping toward evidence-based inventory intelligence.

3 Analytical Architecture

The REEM V2 inventory layer is positioned between the operational data environment and management decision-making.

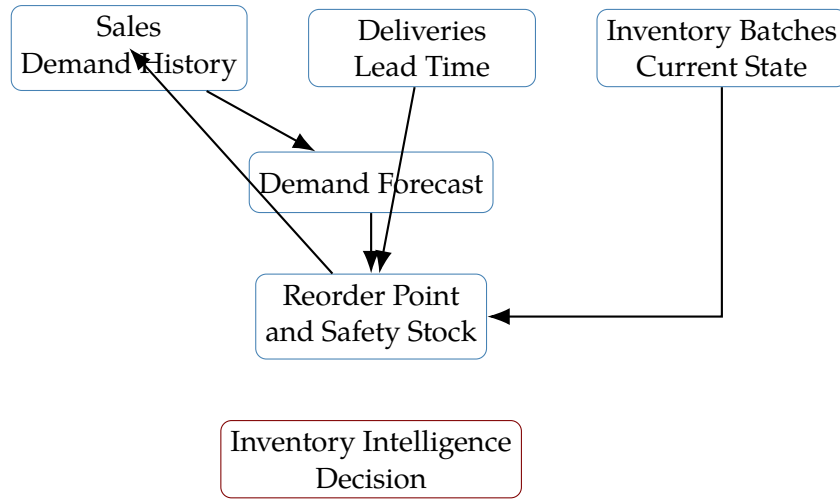


Figure 1: REEM V2 inventory intelligence architecture.

4 Evidence Architecture

A major methodological improvement in REEM V2 is the distinction between transactional evidence and state evidence.

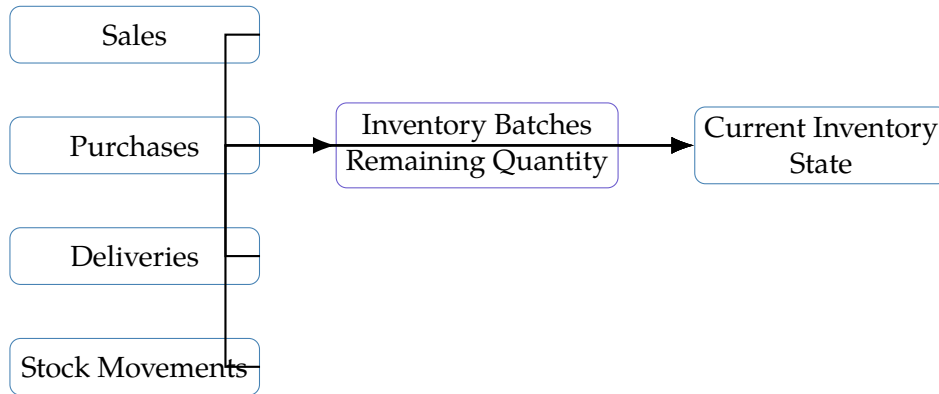


Figure 2: Separation of transactional and state evidence.

This distinction is important.

Stock movements describe inventory events such as sales and receipts. They do not necessarily constitute a complete opening-to-current inventory ledger.

For the present synthetic environment, current stock is therefore derived from:

$$\text{Current Stock}_i = \sum_b \text{RemainingQuantity}_{ib}.$$

This avoids incorrectly producing negative inventory merely because the movement table does not contain a complete opening balance.

5 Demand Input

The inventory model receives expected daily demand from the REEM V2 demand forecasting layer.

For medicine i :

$$\hat{D}_i = \text{Selected Forecast Units per Day.}$$

The current demand model evaluates four development forecasting methods:

1. Naive,
2. seven-day moving average,
3. seven-day weighted moving average,
4. linear trend.

The selected model is based on chronological validation performance.

The inventory model does not independently reinterpret the demand forecast. It consumes the selected forecast as an analytical input.

6 Lead-Time Evidence

Lead time is derived from delivery records whenever delivery history is available.

For medicine i :

$$L_i = \text{observed delivery lead time.}$$

The current development dataset contains:

$$46/50 = 92\%$$

of medicines with observed lead-time evidence.

Four medicines lack sufficient delivery evidence:

- D0015 – Bisoprolol,
- D0023 – Clopidogrel,
- D0004 – Ambroxol,
- D0036 – Insulin.

For these medicines, the development model uses:

$$L_i = 7 \text{ days}$$

with the explicit status:

$$\text{ASSUMED_7_DAYS.}$$

The assumption is deliberately retained in the output rather than being presented as observed evidence.

7 Demand During Lead Time

Expected demand during replenishment lead time is calculated as:

$$D_{LT,i} = \hat{D}_i L_i.$$

For example, D0027 (Enalapril) has:

$$\hat{D} = 7.8571$$

units per day and:

$$L = 9.52$$

days.

Therefore:

$$D_{LT} = 7.8571 \times 9.52 \approx 74.8.$$

This represents expected demand during the replenishment period.

8 Safety Stock

The current development implementation uses validation forecast error as a provisional proxy for demand uncertainty.

The development formulation is:

$$SS_i = 1.65 \times MAE_i \times \sqrt{L_i}.$$

The factor 1.65 corresponds approximately to a one-sided 95% normal-theory service factor. However, this formulation is explicitly provisional.

A production-quality model should eventually estimate uncertainty directly from longitudinal demand residuals and incorporate lead-time variability.

Therefore the current safety-stock calculation should be regarded as:

Development-level safety-stock estimation

rather than a final pharmaceutical inventory policy.

9 Reorder Point

The current reorder point is:

$$ROP_i = D_{LT,i} + SS_i.$$

For D0027:

$$D_{LT} = 74.8$$

and:

$$SS = 9.87.$$

Therefore:

$$ROP = 74.8 + 9.87 = 84.67.$$

The current stock is:

$$11\,007.$$

Thus the available stock is substantially above the calculated reorder threshold.

10 Inventory Decision Logic

The current development decision rule is:

$$\text{Decision}_i = \begin{cases} \text{STOCKOUT_RISK}, & \text{Stock}_i \leq 0 \\ \text{REORDER}, & \text{Stock}_i \leq ROP_i \\ \text{WATCH}, & \text{Stock}_i \leq 1.5ROP_i \\ \text{ADEQUATE}, & \text{Stock}_i > 1.5ROP_i. \end{cases}$$

This is a transparent development rule.

It is intentionally simple and interpretable.

It should not yet be considered a validated operational pharmacy policy.

11 Current Inventory Results

The validated development run produced:

Measure	Result
Medicines analysed	50
Minimum current stock	175
Maximum current stock	11,619
Minimum reorder point	0.77
Maximum reorder point	84.67
Observed lead-time medicines	46
Assumed lead-time medicines	4
Stockout risk	0
Reorder	0
Watch	0
Adequate	50

All medicines therefore satisfy:

$$\text{Stock}_i > ROP_i$$

in the current synthetic environment.

12 Stock-to-Reorder-Point Analysis

The ratio:

$$R_i = \frac{\text{Stock}_i}{ROP_i}$$

provides a simple measure of how far current stock lies above the calculated reorder threshold.

The lowest ten ratios observed were:

Medicine	Stock	Stock/ROP
Cefixime	470	40.1
Omeprazole	526	43.4
Bisoprolol	492	70.4
Doxycycline	686	70.6
Spironolactone	335	85.7
Furosemide	543	90.2
Azithromycin	953	94.8
Ciprofloxacin	805	100.0
Amoxicillin	962	100.1
Captopril	1138	112.3

Even the lowest observed ratio is approximately 40 times the reorder point.

This explains why no medicine is classified as REORDER or WATCH.

13 Interpretation

The result demonstrates that the inventory intelligence pipeline is functionally coherent.

The system successfully connects:

Forecast → Lead Time → Safety Stock → ROP → Current Stock → Decision.

The result also demonstrates an important analytical principle:

The decision engine should describe the data, not manufacture a desired management result.

The synthetic pharmacy contains substantially more stock than is required by the current demand and replenishment assumptions.

Consequently, the correct result is not to artificially create reorder alerts, but to report the observed state:

50 medicines: ADEQUATE

within the current development environment.

14 Evidence Quality

REEM V2 explicitly distinguishes evidence quality.

Component	Evidence	Status
Demand forecast	Historical sales	Development
Lead time	Delivery history	92% observed
Lead time	Missing history	8% assumed
Current stock	Batch remaining quantity	Development
ROP	Calculated	Development
Decision	Rule-based	Development

The evidence ladder is:

Calculated → Analytically Useful → Validated → Operationally Trusted.

The current inventory model is between the first two stages and is being developed toward formal validation.

15 Model Integrity Validation

The current validation produced the following results:

Integrity Test	Result
50 medicines	PASS
No negative stock	PASS
Positive reorder points	PASS
All lead times observed	FAIL
No missing decisions	PASS
Development boundary preserved	PASS
Operational use prohibited	PASS

The single failed test is intentional evidence rather than a hidden error.

Four medicines do not have observed delivery histories. The model therefore correctly identifies their lead time as assumed rather than falsely labeling it as observed.

16 Important Limitations

The current model has several important limitations.

16.1 Synthetic Data

All current records originate from:

SIMULATED_REEM_V2.

They are development records and are not operational evidence.

16.2 Safety Stock

The present safety-stock formulation uses forecast MAE as a proxy for uncertainty.

A future version should use empirical demand variability and lead-time variability.

16.3 Inventory State

Current stock is derived from batch remaining quantities. This is appropriate for the present synthetic state representation but does not replace a fully reconciled production inventory ledger.

16.4 Decision Thresholds

The current decision thresholds are development rules. They require validation against actual pharmacy service levels, stockout costs, criticality, procurement constraints, and management policy.

16.5 Overstocked Synthetic Environment

The current synthetic inventory contains substantially more units than required by modeled demand.

Therefore the present results cannot be used to estimate realistic pharmacy stock requirements.

17 Next Development Stage

The next stage should improve the scientific quality of the inventory model rather than merely add more output.

The development roadmap is:



Figure 3: Proposed inventory intelligence development pathway.

Future extensions may include:

- empirical demand standard deviation,
- lead-time standard deviation,
- service-level targets,
- medicine criticality,
- stockout probability,
- supplier reliability,
- expiry risk,
- ABC classification,
- EOQ,

- Monte Carlo inventory simulation,
- multi-objective inventory optimisation.

18 Management Intelligence Perspective

REEM V2 should ultimately move beyond the question:

“How many units are currently in stock?”

toward questions such as:

- What demand is expected?
- How long will replenishment take?
- How much inventory is required during replenishment?
- How uncertain is the requirement?
- Which medicines are approaching risk?
- Which medicines are excessively stocked?
- Which suppliers create replenishment risk?
- Which inventory should receive management attention first?

This represents the central transition from an operational pharmacy system to a management intelligence system.

19 Conclusion

The REEM V2 inventory intelligence layer has successfully established a coherent development pipeline linking demand forecasting, procurement lead time, safety stock, reorder point, current inventory state, and inventory decision.

The current validation confirms:

- current stock is non-negative,
- reorder points are positive,
- delivery lead times are explicitly distinguished as observed or assumed,
- all 50 medicines receive a decision,
- synthetic-data boundaries are preserved,
- no operational conclusion is claimed.

The model is therefore ready for the next stage of research: improving uncertainty estimation, service-level modelling, stockout risk prediction, supplier intelligence, and inventory optimisation.

REEM V2 Principle

*Operational data should become evidence,
evidence should become intelligence,
and intelligence should support better decisions.*

REEM V2

Research and Development Environment

Data Source: SIMULATED_REEM_V2

Data Status: DEVELOPMENT

Operational Use: PROHIBITED